

ELECTRONIC COMMUNICATION SYSTEMS

NETW310 FINAL COURSE PROJECT

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INTRODUCTION

THIS PROJECT EXPLORES THE FOUNDATIONAL PRINCIPLES OF ELECTRONIC COMMUNICATION SYSTEMS THAT SUPPORT MODERN DIGITAL TRANSMISSION. IT EXAMINES HOW INFORMATION IS REPRESENTED, TRANSMITTED, AND RECEIVED THROUGH BOTH WIRELINE AND WIRELESS MEDIA. EMPHASIS IS PLACED ON UNDERSTANDING HOW MODULATION, ENCODING, AND SIGNAL PROCESSING TECHNIQUES INFLUENCE SYSTEM PERFORMANCE AND RELIABILITY.

KEY TOPICS ADDRESSED IN THIS PROJECT INCLUDE:

BASEBAND AND BANDPASS SIGNAL TRANSMISSION

AMPLITUDE AND FREQUENCY MODULATION (AM/FM)

PULSE CODE MODULATION (PCM) AND LINE CODING TECHNIQUES

ANTENNA GAIN, FREE SPACE PATH LOSS, AND FADING CHANNEL EFFECTS

THROUGH ANALYSIS AND SIMULATION, THIS PROJECT DEMONSTRATES HOW THESE CONCEPTS WORK TOGETHER TO ENABLE EFFICIENT AND RELIABLE COMMUNICATION SYSTEMS.



MODULE 1

COMMUNICATION SYSTEM BASICS

BEFORE ANALYZING MODULATION TECHNIQUES AND TRANSMISSION CHANNELS, THIS PROJECT BEGAN WITH A REVIEW OF ESSENTIAL COMMUNICATION SYSTEM MEASUREMENT TOOLS AND POWER CONCEPTS. MODULE 1 FOCUSED ON UNDERSTANDING HOW SIGNALS ARE OBSERVED IN BOTH THE TIME AND FREQUENCY DOMAINS AND HOW SIGNAL POWER IS MEASURED AND CALCULATED.

KEY AREAS COVERED INCLUDE:

USE OF **OSCILLOSCOPES** TO ANALYZE TIME-DOMAIN SIGNAL BEHAVIOR

USE OF **SPECTRUM ANALYZERS** TO EXAMINE FREQUENCY-DOMAIN CHARACTERISTICS AND BANDWIDTH

REVIEW OF **RF POWER MEASUREMENTS** USING WATTS, DECIBELS (DB), AND DECIBEL-MILLIWATTS (DBM)

APPLICATION OF **3 DB AND 10 DB RULES** FOR BASIC RF POWER CALCULATIONS

This foundational knowledge supports later analysis of modulation, signal transmission, antenna performance, and fading channel behavior.



MODULE 2

BASEBAND SIGNALS AND AM/FM SIGNALS

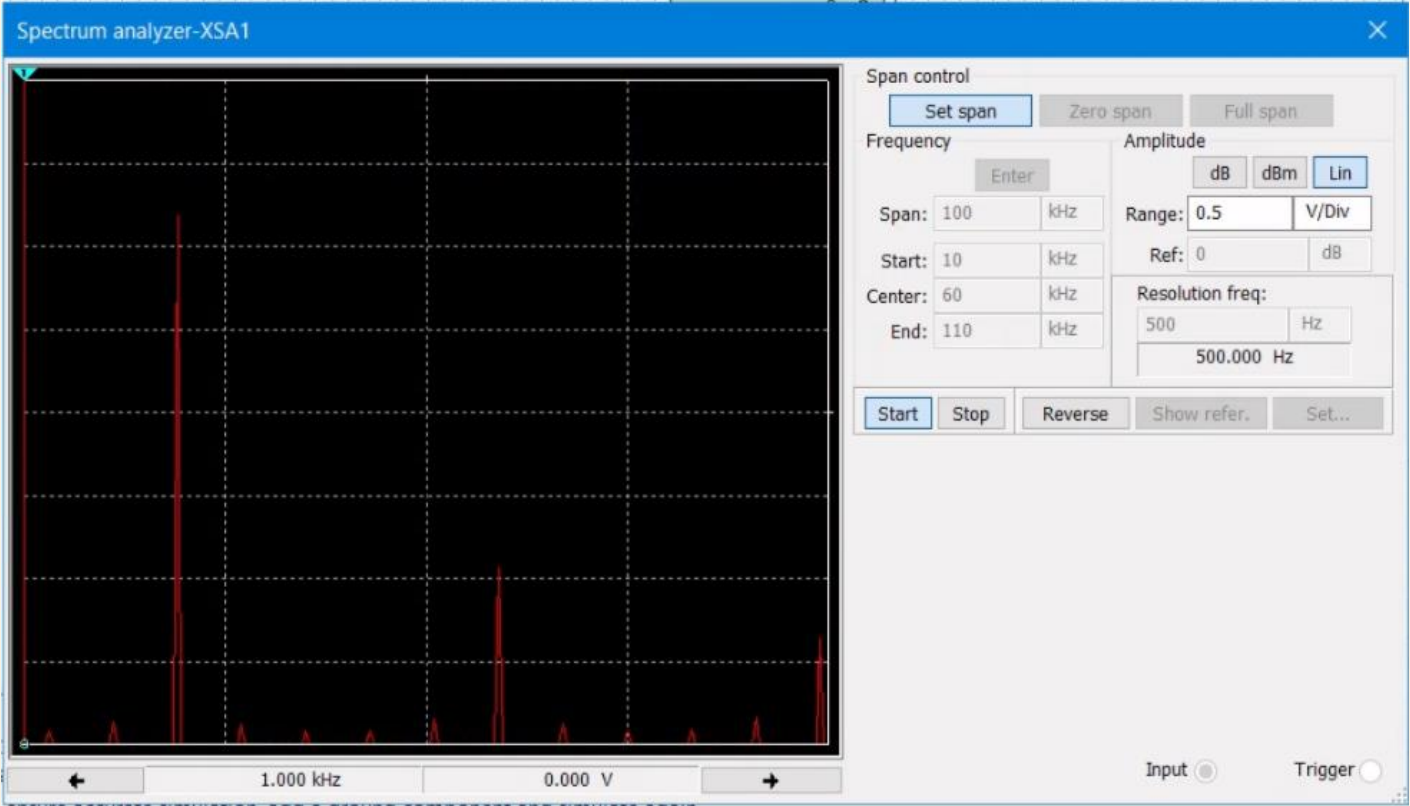
Module 2 applies time-domain and frequency-domain analysis techniques to baseband, AM, and FM signals using Multisim simulations. Oscilloscopes and spectrum analyzers are used to observe square wave harmonics, modulation effects, and signal spectral behavior.



Baseband Signals

Spectrum Analyzer Results

The spectrum analyzer display illustrates the harmonic frequency components of the signal along with the corresponding instrument settings. The signal voltage values at each harmonic frequency are recorded in the table below.



Harmonic Frequency	Signal (Volts)
20 kHz	2.94 V
40 kHz	0 V
60 kHz	0.86 V
80 kHz	0 V
100 kHz	0.59 V

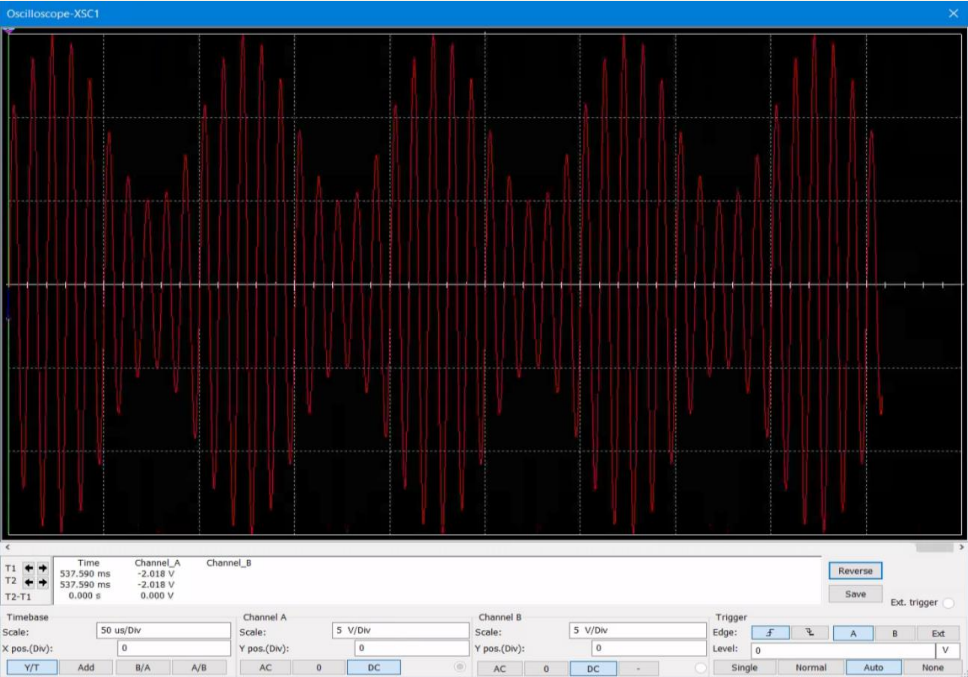
AM Signals

Measurement Results

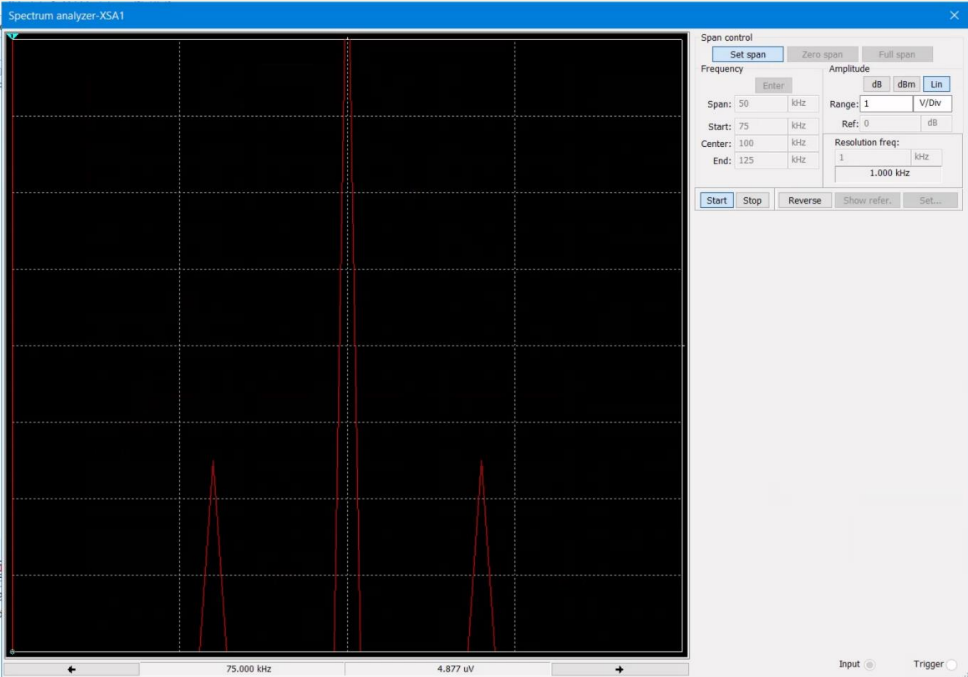
The oscilloscope display shows the signal behavior in the time domain along with the corresponding instrument settings. The spectrum analyzer display presents the signal characteristics in the frequency domain, including the analyzer configuration.

AM Modulation	Frequency (kHz)
Lower side band	90 kHz
Carrier	100 kHz
Upper side band	110 kHz

Oscilloscope



Spectrum Analyzer



AM Signals (Continued)

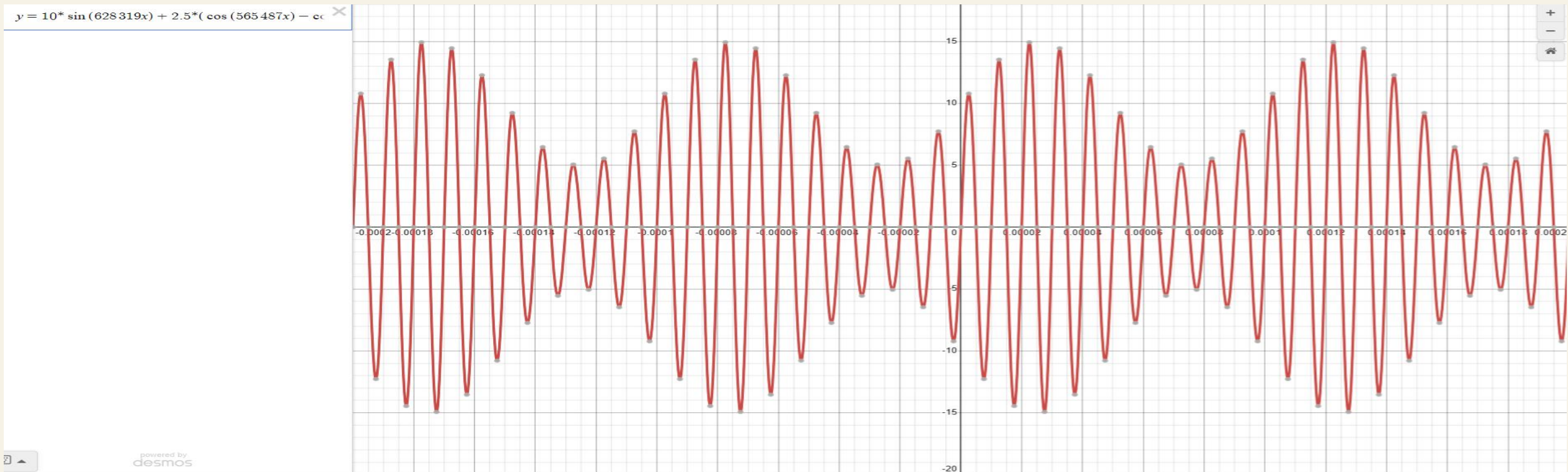
The carrier and modulating signal equations are used to model the amplitude-modulated waveform. These equations illustrate how the information signal varies the amplitude of the carrier signal to produce the AM waveform.

The equation for the carrier signal is: $v_c = 10\sin(2\pi(100000)t)$

The equation for the modulating signal is: $v_i = 5\sin(2\pi(10000)t)$

The equation for amplitude modulated (AM) wave V_{AM} is: $v_{AM} = [10 + 5\sin(2\pi(10000)t)]\sin(2\pi(100000)t)$

The plotted waveform demonstrates the variation of the carrier amplitude in response to the modulating signal.

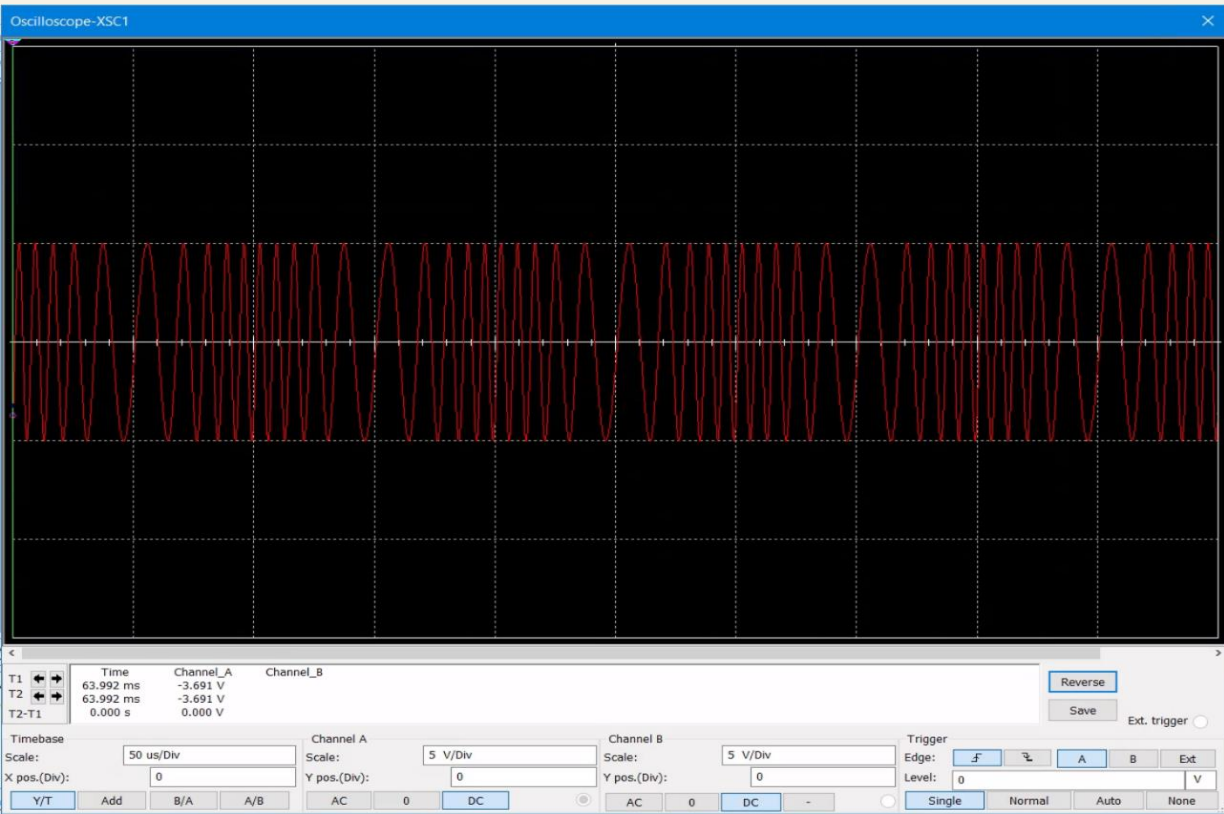


FM Signals

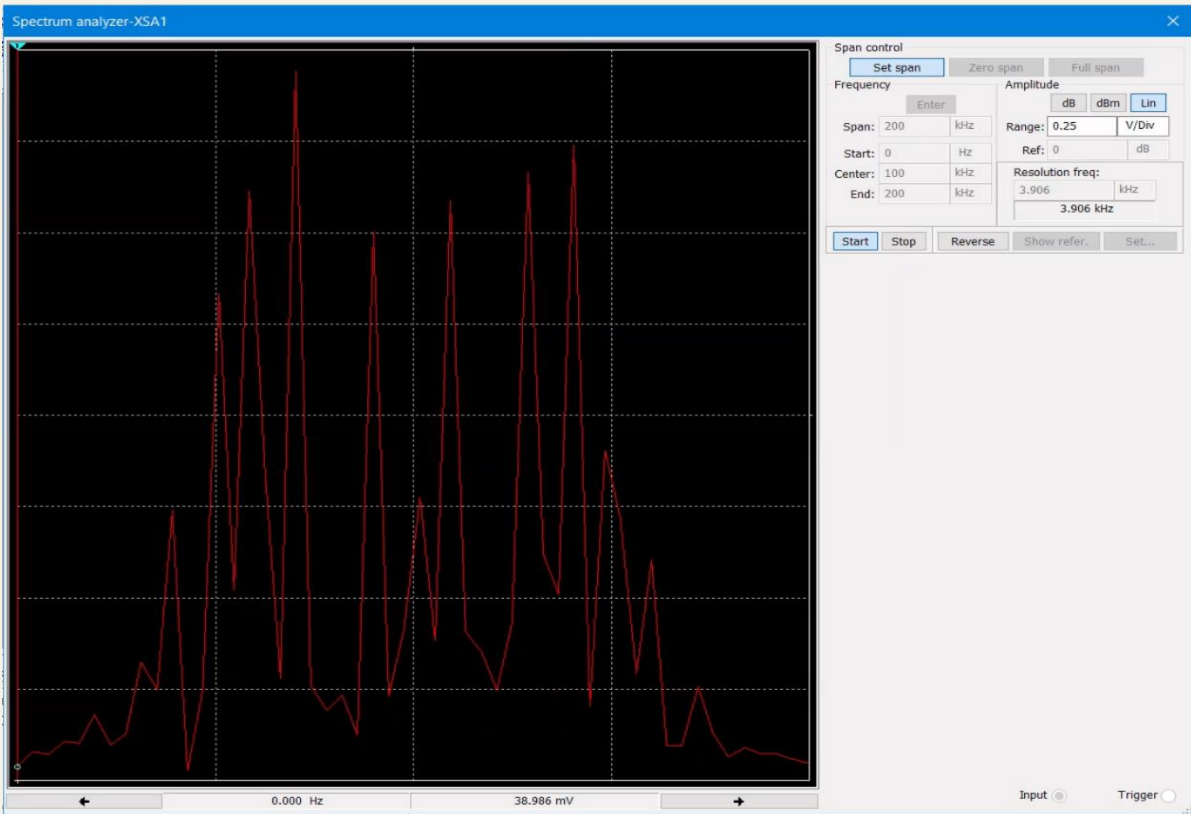
The oscilloscope display shows the FM signal behavior in the time domain along with the corresponding instrument settings. The spectrum analyzer illustrates the signal's frequency-domain characteristics, including the distribution of spectral components and analyzer configuration.

Carrier frequency: 100 kHz

Oscilloscope



Spectrum Analyzer



MODULE 3

PULSE CODE MODULATION (PCM) AND LINE CODES

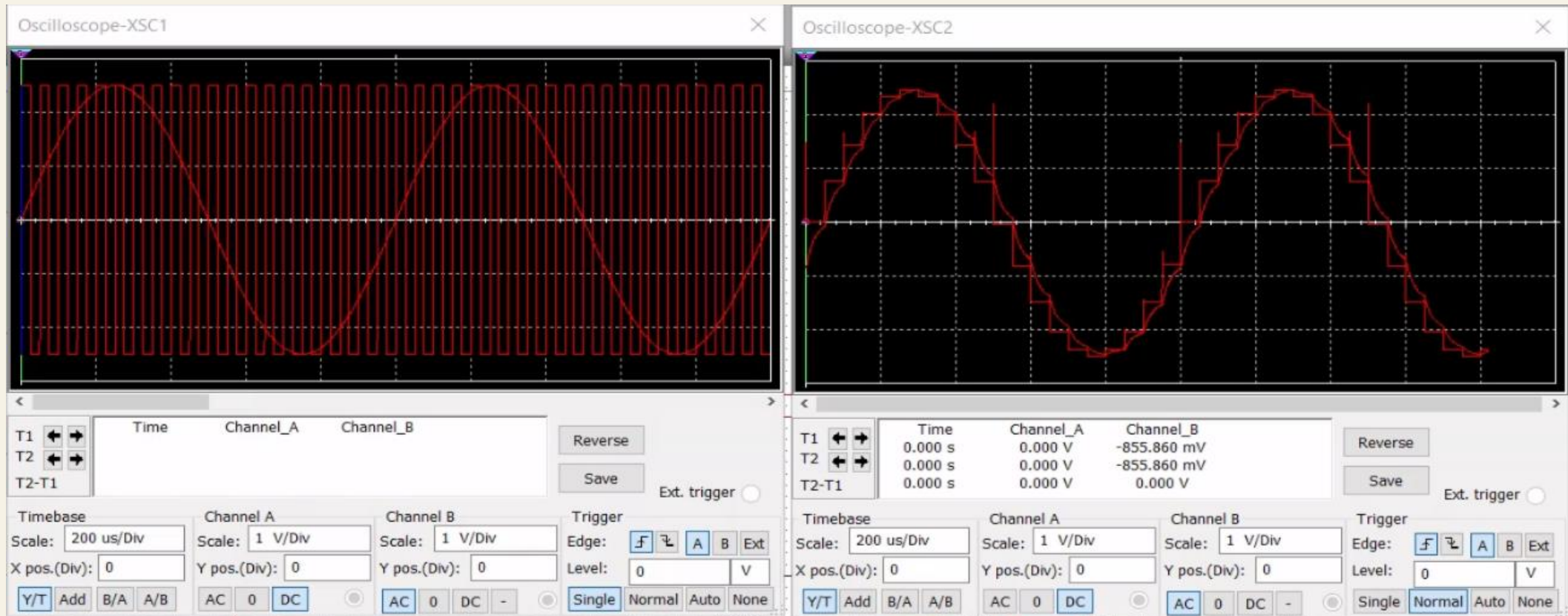
Module 3 explores the basic building blocks of PCM and evaluates the spectral characteristics of common line coding techniques. Simulations are used to compare transmitted and reconstructed signals and to analyze bandwidth requirements.



Pulse Code Modulation (PCM)

The first oscilloscope display shows the analog input signal and the sampling clock at the input of the analog-to-digital converter (ADC). The second oscilloscope display illustrates the output of the digital-to-analog converter (DAC) after signal reconstruction.

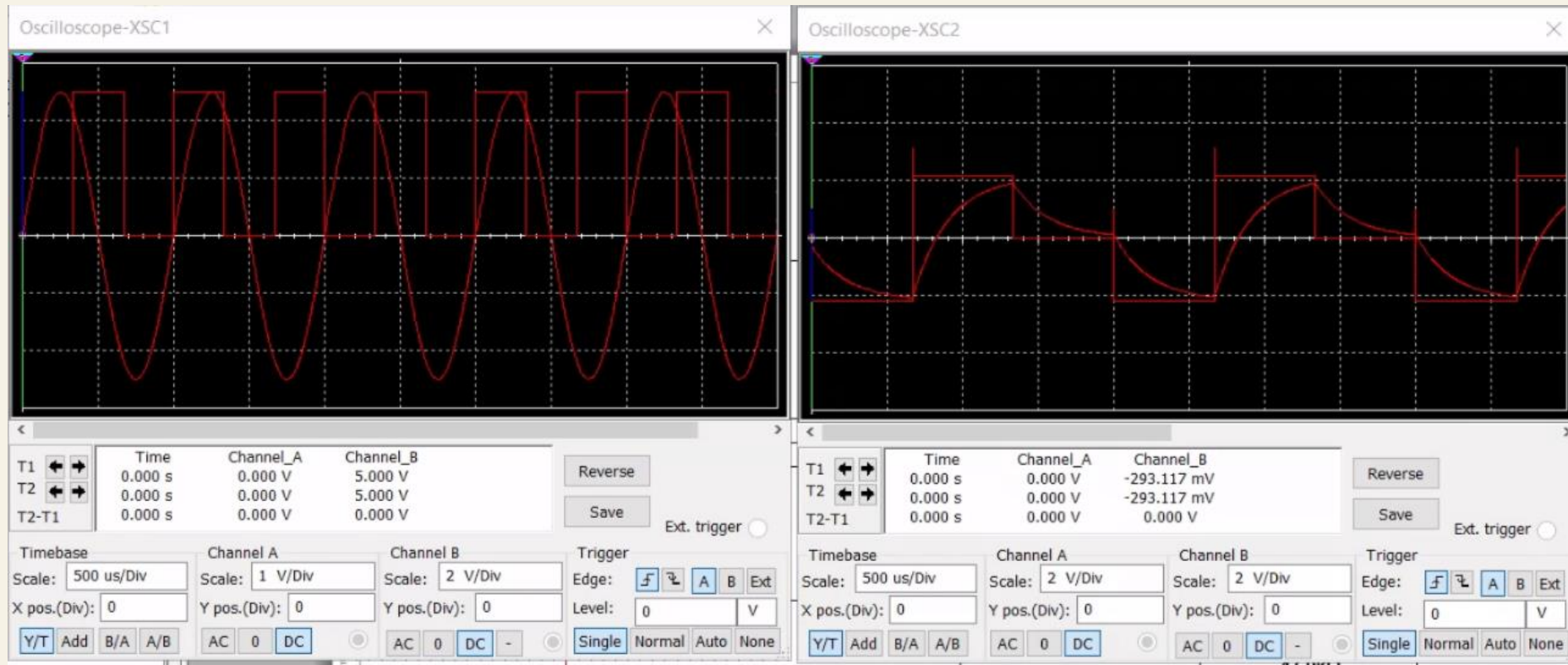
These observations allow for a comparison between the original transmitted signal and the recovered signal at the receiver.



Pulse Code Modulation (PCM)

The first oscilloscope display shows the analog input signal and the sampling clock at the input of the analog-to-digital converter (ADC). The second oscilloscope display illustrates the output of the digital-to-analog converter (DAC) after signal reconstruction.

Increasing the sampling rate improves the accuracy of the reconstructed signal because more samples of the original analog waveform are taken, resulting in a smoother output that closely matches the original signal. Decreasing the sampling rate reduces the number of samples, which causes distortion as important portions of the signal are missed, producing a more blocky and less accurate output.

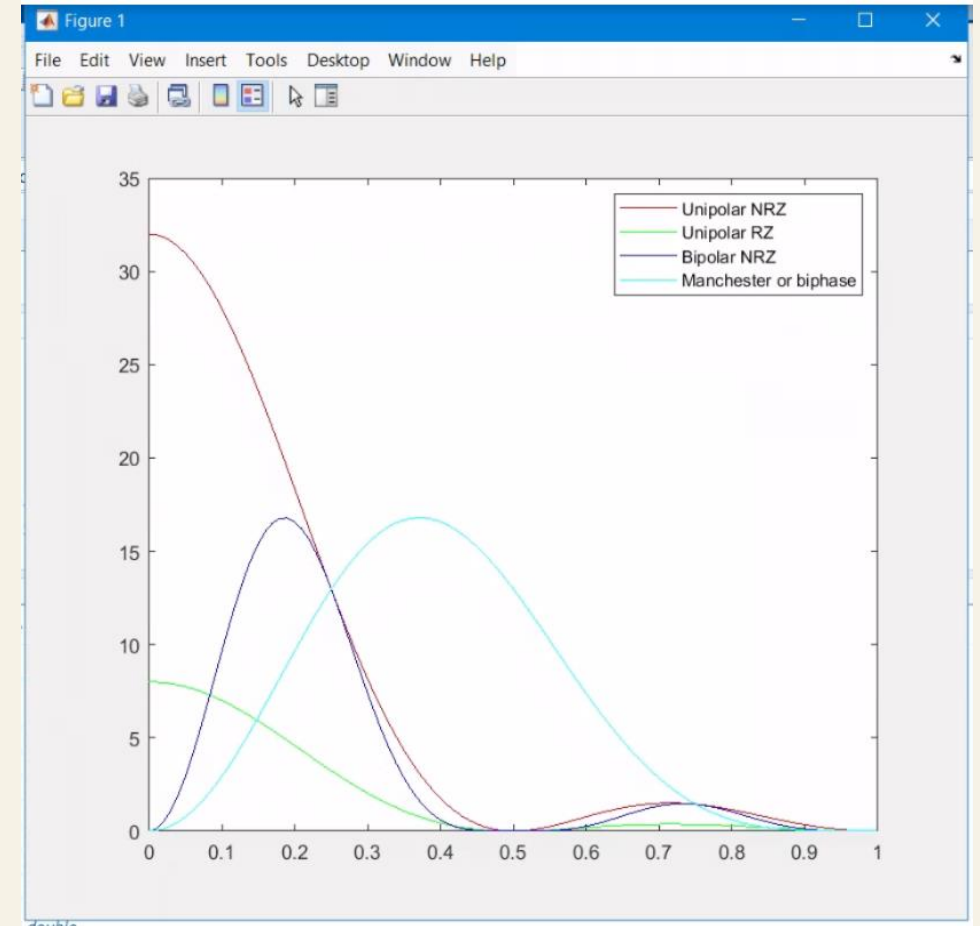


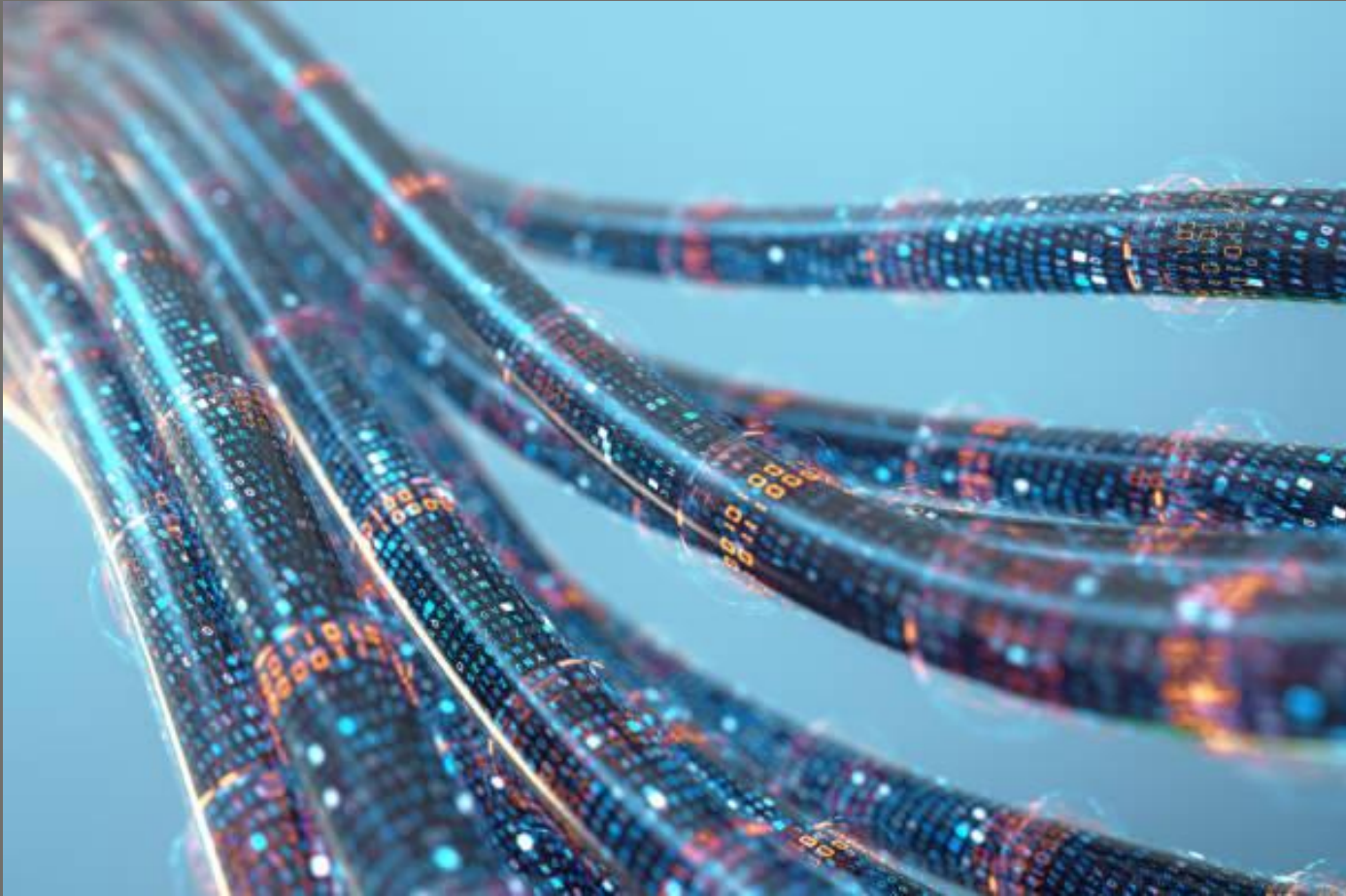
Line Codes

The diagram shows the power spectral densities of four line coding techniques: **NRZ-Polar**, **NRZ-Unipolar**, **NRZ-Bipolar**, and **Manchester-Polar**. These plots illustrate how each line code distributes signal energy across the frequency spectrum.

A comparison of **NRZ-Bipolar** and **Manchester-Polar** encoding shows that Manchester encoding requires more bandwidth. At approximately **0.5 normalized frequency**, the Manchester-encoded signal still maintains significant spectral energy, while the other line codes approach zero. This indicates that Manchester encoding spreads energy over a wider frequency range, resulting in higher bandwidth requirements compared to NRZ-Bipolar and other encoding techniques.

MATLAB Diagram





MODULE 4

CABLES AND STRUCTURED CABLING

Module 4 reviews industry-standard cabling practices for structured cabling systems used in multimedia communications. Standards such as ANSI/TIA-568 and ISO/IEC 11801 are examined to understand proper installation methods, safety requirements, and performance considerations for network cabling.



Questions

1. Why must the twisting in the individual wires be maintained in a UTP cable?

Answer: Twisting must be maintained to reduce electromagnetic interference (EMI) and crosstalk, ensuring proper signal integrity. The Twisting causes external noise to affect both wires equally, allowing it to be canceled out by differential signaling (Frenzel, 2015).

2. How many inches should UTP cable be separated from 110 volts electrical cable?

Answer: UTP cable should be separated by at least 12 inches from 110V electrical cable to reduce electromagnetic interference (EMI), which can distort or degrade data signals (BICSI, 2018).

3. Horizontal cabling connects what areas to each other?

Answer: Horizontal cabling connects the telecommunication room (TR) to the work area (WA) on the same floor, providing a pathway for data communication between network equipment and end devices (Anixter, n.d.).

Questions

4. What is a plenum rated cable?

Answer: A plenum-rated cable is a cable designed for use in air-handling spaces (plenums) that is made of fire-resistant material and does not emit toxic fumes when burned. This presents minimum smoke and toxic fumes in the event of a fire, because these spaces circulate air throughout a building (Frenzel, 2015; BICSI, 2018).

5. What is a riser tube used for?

Answer: A riser tube is used to route cables vertically between floors in a building, supporting backbone cabling and maintaining organized and protected cable pathways. (Anixter, n.d.).

6. Is the grounding of equipment mostly a safety or a performance concern?

Answer: Grounding equipment is mostly a safety concern because it provides a low-resistance path for fault current to flow to earth, preventing electric shock and reducing the risk of fire. Although grounding also helps reduce EMI, its primary purpose is protection (Frenzel, 2015).

MODULE 5

ANTENNA GAIN AND FREE SPACE PATH LOSS

Module 5 explores antenna gain and free space path loss by analyzing how frequency, antenna size, and distance affect wireless signal strength. These principles help explain signal focusing, propagation loss, and overall transmission performance.



Antenna Gain

1. What is the maximum theoretical antenna gain of a common dish antenna at the 2.4 GHz band?

Answer: Maximum theoretical antenna gain at 2.4 GHz band = **22.38 dB**

2. What is the maximum theoretical antenna gain of a common dish antenna at the 5 GHz band?

Answer: Maximum theoretical antenna gain at 5G GHz band = **29.08 dB**

3. Given the same sized reflector, which signals, high-frequency, or low-frequency, can be more efficiently focused by a common dish antenna (i.e., result in a higher antenna gain)?

Answer: High-frequency signals, because they produce higher antenna gain with the same dish size.

4. What is the maximum theoretical antenna gain of the dish antenna used in the VLA radio telescopes in New Mexico at the 5 GHz band?

Answer: The maximum theoretical antenna gain of the dish antenna used in the VLA radio telescope in New Mexico at 5GHz band = **63.06**

5. Given the same signal frequency, which dish antennas, large-sized or small-sized, are more efficient at focusing the signal (i.e., result in a higher antenna gain)?

Answer: Larger-sized dish antennas are more efficient at focusing the signal and result in a higher antenna gain.

Free Space Path Loss

1. What is the free space path loss in dB at the 2.4 GHz band?

Answer: Free space path at 2.4 GHz (100 m) = 80.4 dB

2. What is the free space path loss in dB at the 5 GHz band?

Answer: Free space path loss at 5 GHz (100 m) = 87.2 dB

3. How does the free space path loss at a higher frequency (e.g., the 5 GHz band) compare with that at a lower frequency (e.g., the 2.4 GHz band)?

Answer: Free space loss increases with frequency; therefore, higher-frequency signals (such as 5 GHz) experience greater loss than lower-frequency signals (such as 2.4 GHz) over the same distance.

4. What is the free space path loss in dB over 20 meters at the 2.4 GHz band?

Answer: 20 m: 73.22 dB

5. What is the free space path loss in dB over 40 meters at the 2.4 GHz band?

Answer: 40 m: 79.24 dB

6. What is the free space path loss in dB over 80 meters at the 2.4 GHz band?

Answer: 80 m: 85.26 dB

7. When the distance doubles, how does free space path loss in dB change approximately?

Answer: When the distance doubles, the free space path loss increases by approximately 6 dB.

Free Space Path Loss Cont.

8. Use a scientific calculator to calculate Delta for $D1 = 20$ meters and $D2 = 40$ meters.

$$\text{Delta} = \underline{\underline{6.02 \text{ dB}}}$$

9. Is your calculation approximately the same as the result from Part 1 Step 2?

Answer: Yes, the calculation change in free space path loss (approximately 6.02 dB) matches the difference observed in Part 1 Step 2 when the distance doubled from 20 m to 40 m.

MODULE 6

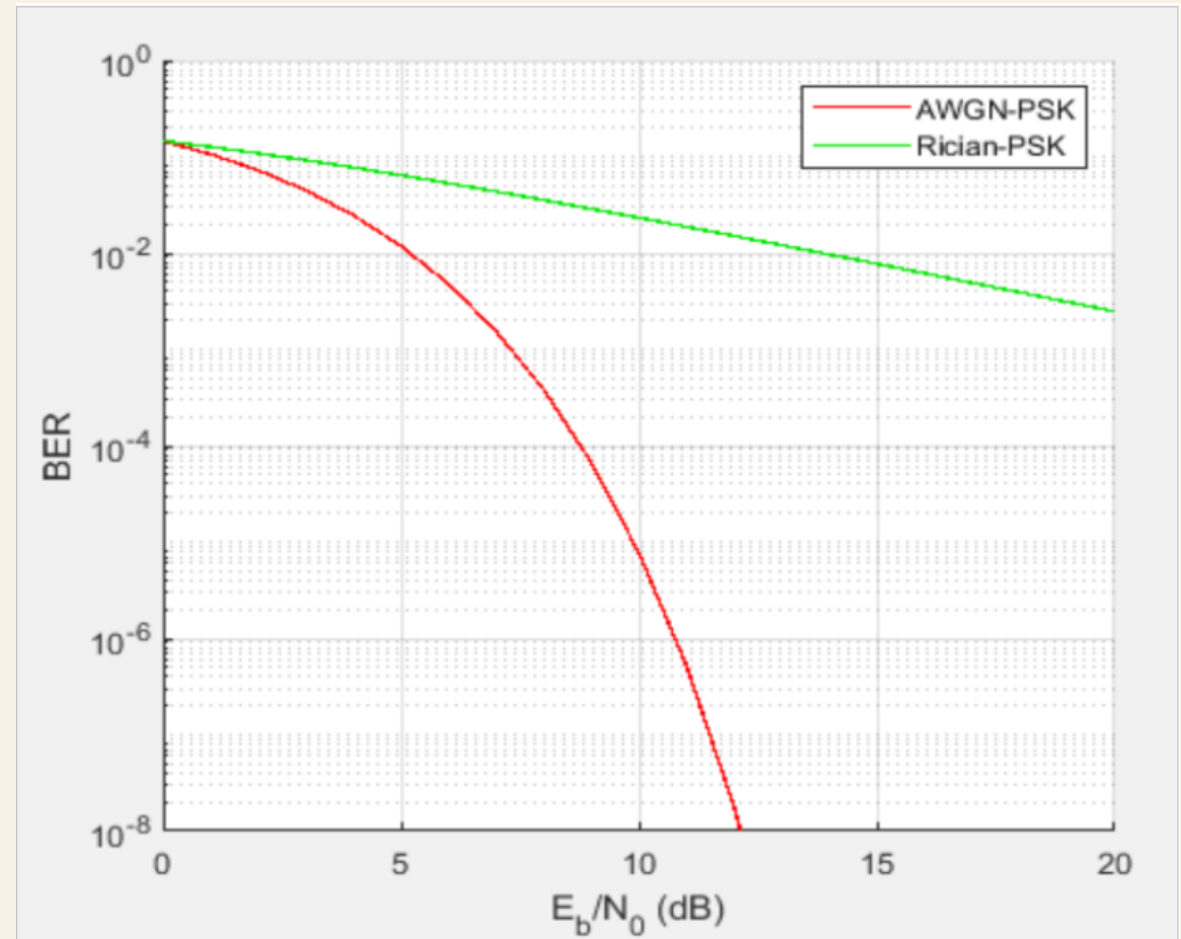
BIT ERROR RATE OF FADING CHANNELS

Module 6 analyzes the BER performance of digital communication systems operating over noise-limited and fading channels. Different modulation schemes and channel conditions are compared to evaluate how noise and fading affect transmission reliability.



AWGN Channel and Rician Channel

The plot illustrates the **bit error rate (BER) performance** of a PSK modulation scheme operating over an **additive white Gaussian noise (AWGN)** channel and a **Rician fading** channel as a function of E_b/N_0



AWGN Channel and Rician Channel

Compare and contrast the BER performance of PSK over a noise limited (AWGN) transmission channel to the performance of PSK over a Rician channel at 5- and 10-dB Signal to Noise Ratio (E_b/N_0), respectively.

Is there a significant difference in BER performance of PSK between the AWGN channel and Rician channel?

Answer:

At 5 dB, the BER for AWGN-PSK is approximately 10^{-2} , while the BER for Rician-PSK is about 6×10^{-2} . This means the Rician channel has several times more bit errors than the AWGN channel at this signal-to-noise ratio.

At 10 dB, the BER for AWGN-PSK decreases to approximately 10^{-5} , while the BER for Rician-PSK remains around 2×10^{-2} . This shows that the AWGN channel performs dramatically better, and the performance gap increases as SNR increases.

In comparison, an AWGN channel is primarily affected by random noise, so increasing SNR significantly improves BER performance. In contrast, a Rician channel is affected by both noise and fading, causing the BER to improve more slowly. Overall, the AWGN channel provides significantly better BER performance than the Rician channel at both 5 dB and 10 dB, with the difference becoming especially large at higher SNR levels.

CHALLENGES AND LESSONS LEARNED

Throughout this course project, several challenges were encountered while analyzing and simulating electronic communication systems. One of the primary challenges was configuring simulation tools correctly, such as oscilloscopes and spectrum analyzers, to obtain accurate time-domain and frequency-domain results. Interpreting spectral plots and understanding how parameter changes affected signal behavior also required careful analysis.

This project reinforced several important lessons:

- Gained a deeper understanding of how **modulation, encoding, and channel conditions** impact system performance
- Developed stronger skills in using **simulation software** to analyze communication signals
- Improved ability to interpret **time-domain, frequency-domain, and BER performance plots**
- Learned how theoretical concepts such as **sampling, bandwidth, antenna gain, and path loss** translate into practical system behavior

Overall, this project strengthened both theoretical knowledge and practical analysis skills, providing valuable insight into the design and evaluation of modern communication systems.



CAREER SKILLS

This course helped me develop practical skills in network signal analysis, modulation, encoding, and performance evaluation. These competencies directly support my career in cybersecurity and networking by strengthening my ability to analyze, troubleshoot, and secure communication systems.



CONCLUSION

THIS COURSE PROJECT PROVIDED A COMPREHENSIVE UNDERSTANDING OF ELECTRONIC COMMUNICATION SYSTEMS AND THE PRINCIPLES THAT SUPPORT MODERN DIGITAL TRANSMISSION. BY EXPLORING MODULATION TECHNIQUES, ENCODING METHODS, TRANSMISSION MEDIA, AND CHANNEL EFFECTS, I GAINED INSIGHT INTO HOW INFORMATION IS REPRESENTED, TRANSMITTED, AND RECOVERED IN BOTH WIRELINE AND WIRELESS ENVIRONMENTS.

THROUGH SIMULATION AND ANALYSIS, THIS PROJECT DEMONSTRATED THE PRACTICAL IMPACT OF SYSTEM DESIGN CHOICES ON SIGNAL QUALITY, BANDWIDTH REQUIREMENTS, AND TRANSMISSION RELIABILITY. TOPICS SUCH AS AM/FM MODULATION, PCM, LINE CODING, ANTENNA GAIN, FREE SPACE PATH LOSS, AND BER PERFORMANCE REINFORCED THE CONNECTION BETWEEN THEORETICAL CONCEPTS AND REAL-WORLD COMMUNICATION BEHAVIOR.

OVERALL, THIS PROJECT STRENGTHENED MY TECHNICAL FOUNDATION IN COMMUNICATIONS AND ENHANCED SKILLS THAT DIRECTLY SUPPORT MY CAREER IN **CYBERSECURITY AND NETWORKING**, PARTICULARLY IN ANALYZING, TROUBLESHOOTING, AND SECURING COMMUNICATION SYSTEMS.



REFERENCES

FRENZEL, L. E. (2015). *PRINCIPLES OF ELECTRONIC COMMUNICATION SYSTEMS* (4TH ED.). MCGRAW-HILL EDUCATION.

MESSAOUD, L. (2026). *NETW310 MODULE 3: PULSE CODE MODULATION AND LINE CODES* [LECTURE RECORDING]. DEVRY UNIVERSITY.

MESSAOUD, L. (2026). *NETW310 MODULE 6: COMMUNICATION TEST AND MEASUREMENTS* [LECTURE RECORDING]. DEVRY UNIVERSITY.

NETW310 COURSE PROJECT GUIDE MODULE 5. (N.D.). *ANTENNA GAIN AND FREE SPACE PATH LOSS*. DEVRY UNIVERSITY.

NETW310 COURSE PROJECT GUIDE MODULE 6. (N.D.). *BIT ERROR RATE OF FADING CHANNELS*. DEVRY UNIVERSITY.

ANIXTER. (N.D.). *ANSI/TIA/EIA-568-B STRUCTURED CABLING STANDARD*.
[HTTPS://WEB.ANIXTER.COM/AXECOM/AXEDOCLIB.NSF/\(UNID\)/8F2E0839A6190F4986257309005757CC/\\$FILE/ANSI-TIA-EIA-568-B.PDF](HTTPS://WEB.ANIXTER.COM/AXECOM/AXEDOCLIB.NSF/(UNID)/8F2E0839A6190F4986257309005757CC/$FILE/ANSI-TIA-EIA-568-B.PDF)

BICSI. (2018). *NEW CABLING STANDARDS FOR THE BUILDINGS OF TODAY AND TOMORROW*.
<HTTPS://WWW.BICSI.ORG/DOCS/DEFAULT-SOURCE/CONFERENCE-PRESENTATIONS/2018-FALL-CONFERENCE/NEW-CABLING-STANDARDS-FOR-THE-BUILDINGS-OF-TODAY-AND-TOMORROW.PDF>

FS COMMUNITY. (N.D.). *NETWORK CABLE STANDARDS: TIA-568 VS ISO/IEC 11801 VS EN 50173*.
<HTTPS://COMMUNITY.FS.COM/BLOG/NETWORK-CABLE-STANDARDS-TIA-568-VS-ISO-11801-VS-EN-50173.HTML>

DESMOS. (N.D.). *DESMOS SCIENTIFIC CALCULATOR*. RETRIEVED APRIL 3, 2026, FROM
<HTTPS://WWW.DESMOS.COM/SCIENTIFIC>

